

L04 Assignment 1 - Brokenmodel

Intro

I have first corrected the 'Brokenmodel.mdl' according to the assignment in Lecture 03.

During the process, I recorded the 'L03 - Notes and Changes' and created the model 'L03_Brokenmodel_Minardi.mdl'. This work corresponds to the points 1.a and 1.b.i of this assignment.

For rest of this assignment starting from point 1.b.i.1, I recorded my observations and changes below at 'L04 - Notes and Changes'. This and created the model 'L04_Brokenmodel_Minardi.mdl'.

L03 - Notes and Changes

Ref.: L03_Brokenmodel_Minardi.mdl

1. There are 11 unit errors
2. Deer stock has no units
3. No link between 'Food consumption rate' and 'food consumption per deer'
4. No link between 'Deer predation rate' and 'Predator'
5. Variables generally do not follow the standard naming convention (lowercase)
6. Constants do not follow the standard naming convention (uppercase)
7. 'predator net increase' unit is not in unit predator/year
8. Renamed 'predator net increase' into 'predator change'
9. Renamed 'predator fractional increase' into 'predator fractional change'
10. No units specified for - Deer kills per predator
11. No units specified for - Food regeneration time
12. Renamed 'Calories per acre' into CALORIES PER ACRE (constant)
13. Renamed 'Fd dr r' into 'food per deer rate'
14. Added 'predator density'
15. Adjusted units for 'deer predation rate'
16. Adjusted 'deer predation rate' in function of 'predator density'
17. Run 'Units Check': shows just warnings for the lookup functions
18. Run 'Check Model': 'Model is OK.'
19. Run the model: model runs and generates results

L04 - Notes and Changes

Ref.: L04_Brokenmodel_Minardi.mdl

Point 1.b.i.1 - Does this uncover any issues?

1. I think that the constant AREA is perhaps unnecessary, for the curve in 'deer kills per predator' is almost linear. I.e. we could define 'deer kills per predator' as a constant (linear relation) and maintain the same trend.
In doing so, we could get rid of 'predator density', for it was basically just needed for normalizing the units in 'deer predation rate' - which would become: deer kills per predator * Predator.
To keep things simple, I removed the variable 'predator density' (added in 14.) and adjusted the units in 'deer kills per predator' and 'deer predation rate'.
2. The main issue that I see is that, basically, the model 'L03_Brokenmodel_Minardi.mdl' does not shows the interrelations in the dynamics of the three populations, as one would expect from a typical predator/pray model (more at point 1.d.i).

1.c - Identify Issues in Presentation of the Model

- I. Generally, the Brokenmodel.mdl does not follow the standard naming conventions.
 - A. There are constant variables that are not highlighted as such (e.g. by writing them in uppercase, or with a color mark).
 - B. There is a variables with capitalised name that is not a constant.
 - C. Variables have first letter capitalised
 - D. No convention for stock naming and level unit names (Some stocks are named singular and others plural, same for their levels)
- II. Predator stock and flows are displayed vertically (perhaps makes the model representation chaotic).
- III. Flows are described with the word 'increase', although the flow direction is not always clear.

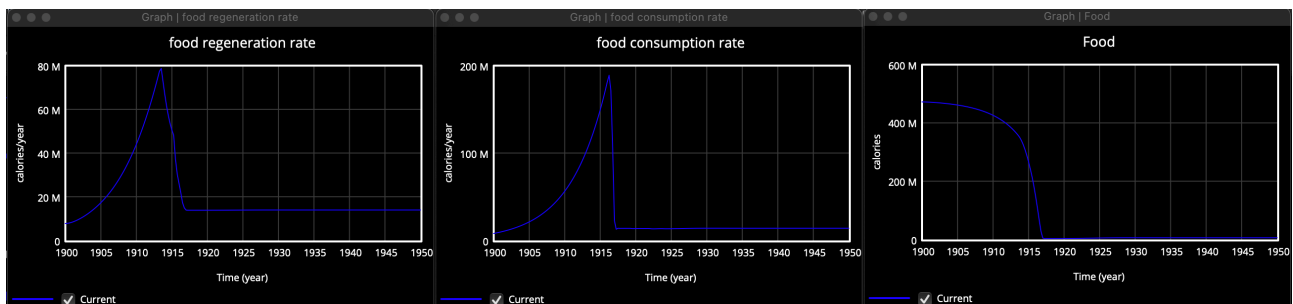
1.d.i Run The model - Can you Explain the Initial Conditions?

Ref.: 03_Brokenmodel_Minardi.mdl

1. Food

Food at t1 is equal to $\text{AREA} \times \text{CALORIES PER ACRE} = 800000 \times 590 = 472'000'000$.

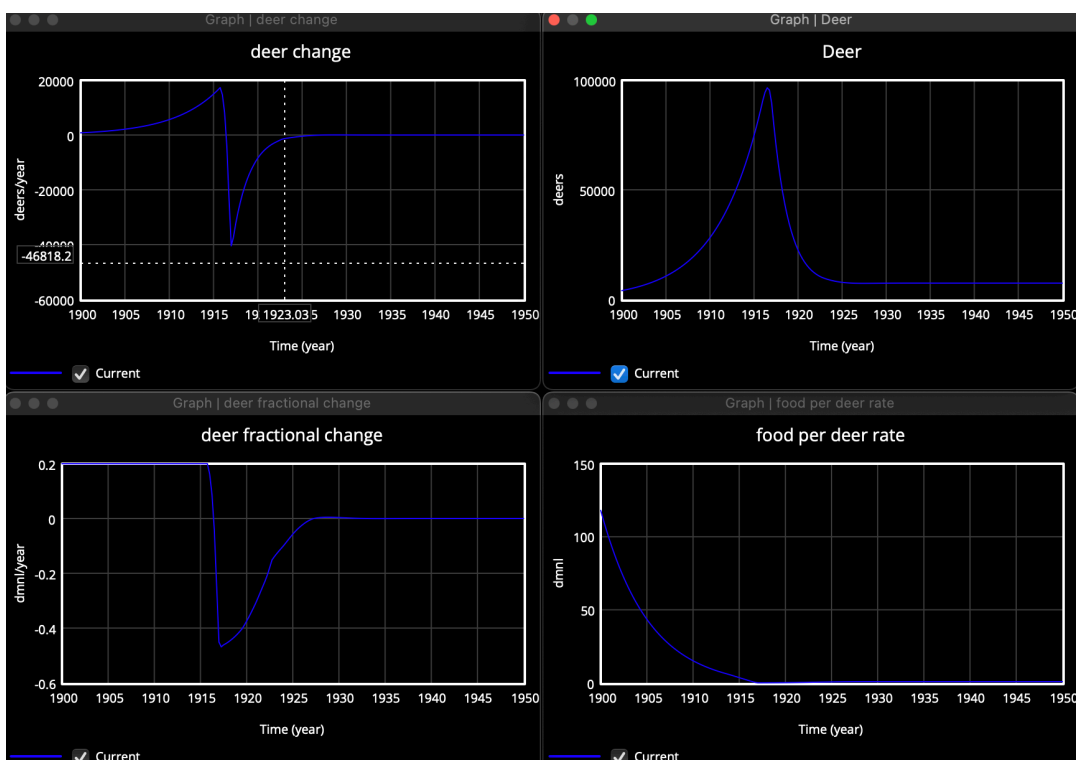
In the first cycles, the food regeneration rate grows a much slower pace than the food consumption rate. The food stock therefore depletes exponentially.



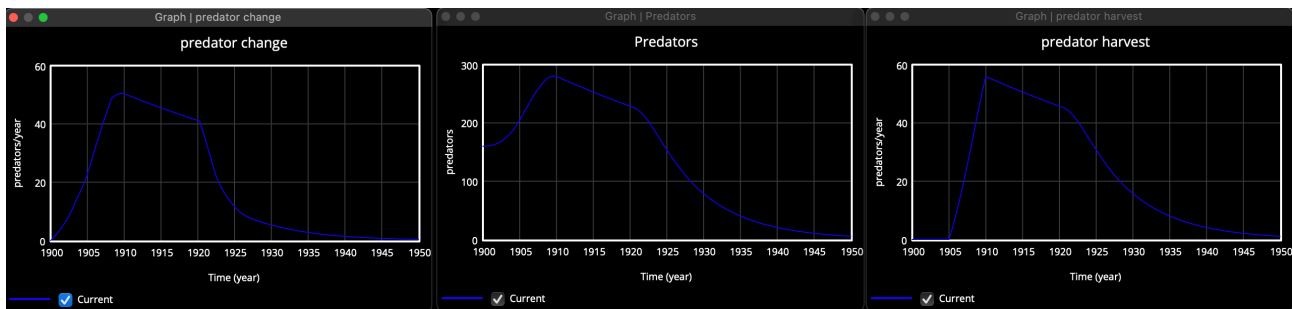
2. Deer

Initial deer population = 4000.

Given food abundance, the deer population increases exponentially (high food per deer rate -> high deer fractional change -> constant growth in deer change).



3. Predator
Initial value = 160
The predator change (inflow) increases earlier than predator harvest. Around 1910 the predator harvest slightly exceeds the predator change and drives the stock down.

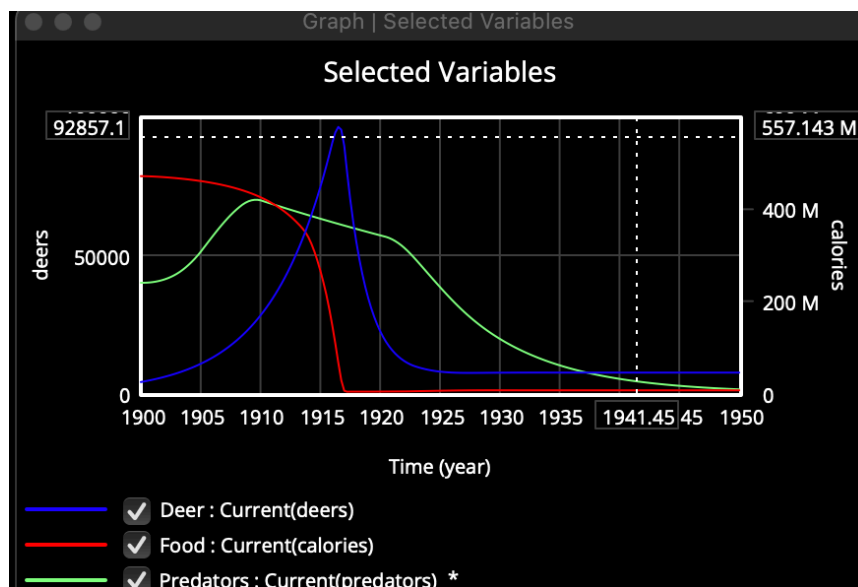


1.d.ii Run The model - Can you get it to Oscillate Making Reasonable Changes?

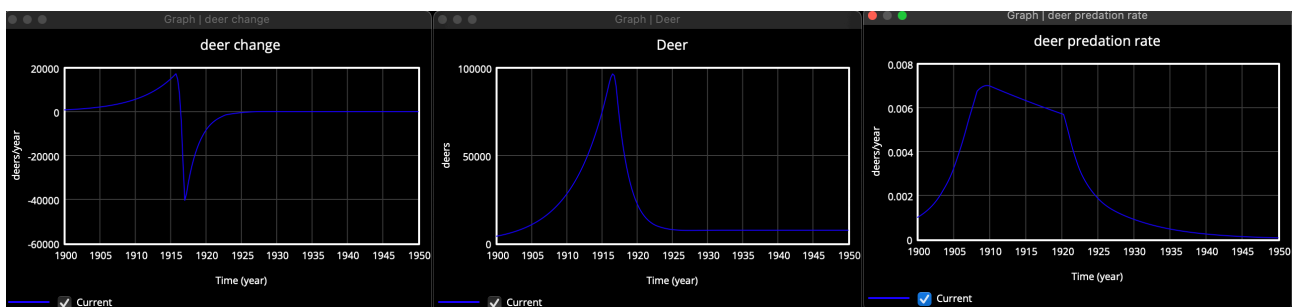
1. Initial situation

Ref.: 03_Brokenmodel_Minardi.mdl

Looking at the stocks, we have these dynamics:



The dynamics displayed are more proper of a 'overshoot and collapse' scenario: the food supply deteriorates exponentially, as the deer population increases exponentially.

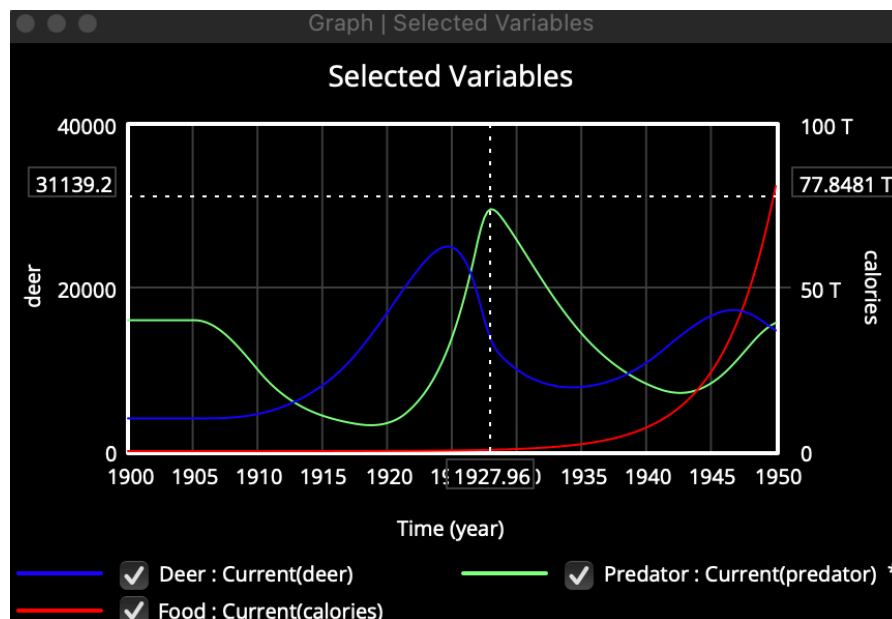


The drop in food supply drives a major drop in deer change. In combination with the relatively high deer predation rate, the deer stock plunges faster than its increase speed. The low deer stock drives down the predator change that cannot compensate for the predator harvest. Therefore The predator stock declines slowly. Given that the speed of food regeneration is too slow after the collapse for pushing the deer population up again, the deer stock remains constant thereafter.

2. Oscillating Scenario

2.1 V1

Ref. L04_Brokenmodel_Minardi.mdl



I have basically adjusted the lookup functions for deer kills per predator (now generally higher) and predator fractional change (higher for high deer kills per predator)

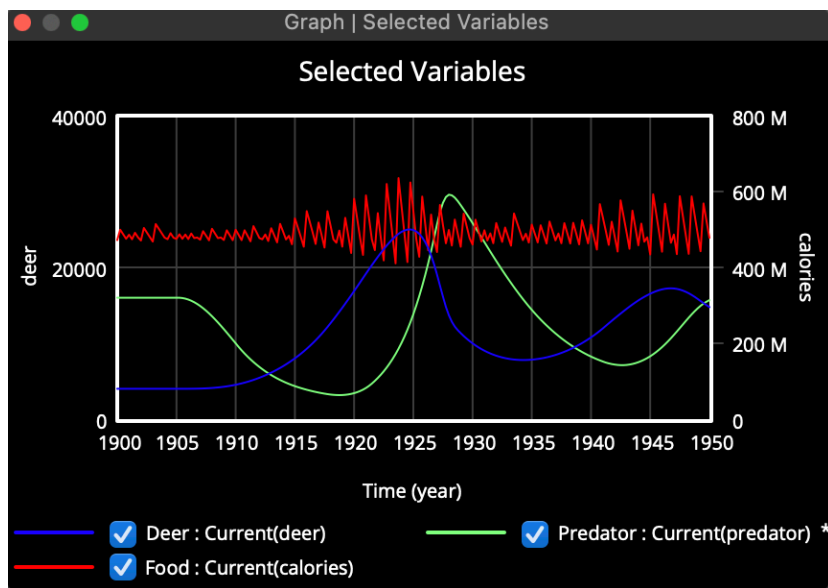
I have done changes in food regenerating and food consumed per deer, but I will discard them in and apply other assumptions in V2.

2.2 V2

Ref. L04_Brokenmodel_Minardi_v2.mdl

In V2:

- * the food consumed per deer is linearly increasing and reaches a higher upper bound - I.e. we have greedy deers that eat more food when there is plenty.
- * The food regeneration rate is set higher for lower values of Food/FOOD MAX.Result:

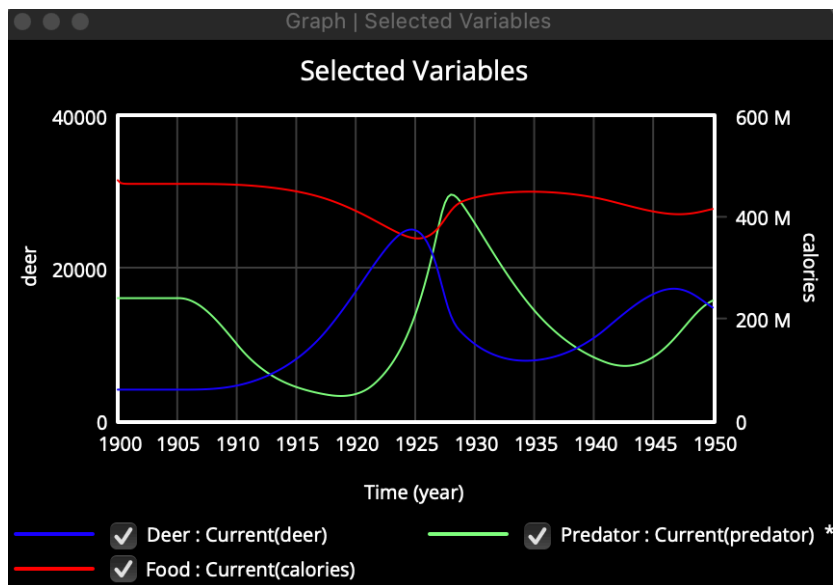


Result:

The food supply remains constant in trend. However it oscillates more when the deer population is higher.

2.3 V3

Ref.: L04_Brokenmodel_Minardi_v3.mdl



By adapting the lookup function for the food regeneration rate again by giving it a more exponential shape, but still expressing it as an inverse relation, we can increase the fluctuation of the food stock and get rid of the oscillations.